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Testing Equality of Variance of population.

Suppose we have two populations, having true mean and true variance, then we computed

For pop 1

N_1, S_1^2, $\bar{X}_1$, σ_1^2, μ_1

For pop 2

N_2, S_2^2, $\bar{X}_2$, σ_2^2, μ_2

Hypothesis:

$H_0 : \sigma_1^2 = \sigma_2^2$ & $\sigma_1^2 - \sigma_2^2 = 0$

① $- H_1 : \sigma_1^2 \neq \sigma_2^2$ & $\sigma_1^2 - \sigma_2^2 \neq 0$

② $- H_1 : \sigma_1^2 > \sigma_2^2$ & $\sigma_1^2 - \sigma_2^2 > 0$

③ $- H_1 : \sigma_1^2 < \sigma_2^2$ & $\sigma_1^2 - \sigma_2^2 < 0$

F-Test:

To compute this, we will use F-test. Assuming population is following normal distribution.

- $F = \frac{S_1^2}{S_2^2}$

follows F-distribution with $V_1 = n_1 - 1$ and $V_2 = n_2 - 1$.

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Critical Region:

For case 1

$$F > F_{-\alpha/2}(V_1, V_2)$$

OR

$$F < F_{1-\alpha/2}(V_1, V_2)$$

For case 2

$$F > F_{\alpha}(V_1, V_2)$$

For case 3

$$F < F_{1-\alpha}(V_1, V_2)$$

QUESTION:

A company wants to test whether two different machines produce products with the same variance in their output measurements. A random sample of **12 observations** was taken from Machine 1, and the standard deviation was found to be **4**. Another random sample of **10** observations was taken from Machine 2, and its standard deviation was **5**.

Assuming that the variances of the two machines are equal, test the hypothesis at a **0.1 significance level**.

Perform an **F-test for equality of variances** and determine whether the null hypothesis should be accepted or rejected. Show all steps and calculations clearly.

Example:

Data:

$N_1 = 12 \quad , \quad S_1 = 4$

$N_2 = 10 \quad , \quad S_2 = 5$

Assume

$\sigma_1^2 = \sigma_2^2$

$\alpha = 0.1$

=> Hypothesis

$H_0 : \sigma_1^2 = \sigma_2^2$

$H_1 : \sigma_1^2 \neq \sigma_2^2$

Test Statistics:

$$F = \frac{S_1^2}{S_2^2} = \frac{(4)^2}{(5)^2} = 0.64$$

Critical Region

$V_1 = 11 \quad , \quad V_2 = 9$

Values at $F_{0.05}(11, 9)$

$$= \frac{3.14 + 3.07}{2} = 3.105$$

Value at $F_{0.95}(11, 9)$

$$F_{1-\beta}(V_1, V_2) = \frac{1}{F_\beta(V_2, V_1)}$$

$$= \frac{1}{F_{0.05}(9, 11)} = \frac{1}{2.90} = 0.344$$

$F > 3.11 \quad \text{OR} \quad F < 0.34$

Claim H_0 Accepted.